

COMPUTER EXERCISE 4 VALENCIA 2019

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1. EXERCISE 1

Run the code given in the exercise

```
> library(gap)
> data(l51, package="gap")
> library(kinship2)
> ped <- with(l51, pedigree(id,fid,mid,sex))
> plot(ped)
> km <- kin.morgan(l51)
> A <- km$kin.matrix*2
> ##### hglm analysis #####
> library(hglm)
> obs.qt <- !is.na(l51$qt) #Remove missing data
> L <- t( chol( A ) )[ obs.qt, ] #Compute
> X <- model.matrix( qt ~ 1, data = l51)
> y <- l51$qt[ obs.qt ]
> mod1 <- hglm(y = y, X = X, Z = L)
```

NOTE: You are fitting a model with one Gaussian random effect term, and the number of effects (p) is greater than the number of observations (n). Consider turning on the argument 'bigRR' that may speed up a lot if $p \gg n$.

```
> summary(mod1)
```

Call:

```
hglm.default(X = X, y = y, Z = L)
```

```
-----
MEAN MODEL
-----
```

Summary of the fixed effects estimates:

Date: May 20, 2019.

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	0.4666	0.2161	2.159	0.0393 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: P-values are based on 29 degrees of freedom

Summary of the random effects estimates:

	Estimate	Std. Error
1	-0.8546	0.4100
2	-0.0505	0.4100
3	-0.3278	0.4153

...

NOTE: to show all the random effects, use `print(summary(hglm.object), print.ran`

DISPERSION MODEL

NOTE: h-likelihood estimates through EQL can be biased.

Dispersion parameter for the mean model:

[1] 0.4424555

Model estimates for the dispersion term:

Link = log

Effects:

	Estimate	Std. Error
	-0.8154	0.2614

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

Dispersion parameter for the random effects:

[1] 0.2847

Dispersion model for the random effects:

Link = log

Effects:

```
.|Random1
  Estimate Std. Error
    -1.2562    0.3963
```

Dispersion = 1 is used in Gamma model on deviances to calculate the standard error

EQL estimation converged in 7 iterations.

Now we check the identity $A = LL$ using all the data

```
> L.all <- t( chol( A ) )
> round( sum( abs( A - tcrossprod(L.all) ) ), 6 )
[1] 0
```

2. EXERCISE 2

The variance for the random effects is 0.2847 and the residual variance is 0.4425. The heritability is consequently $0.2847/(0.2847 + 0.4425)$.

3. EXERCISE 3

The breeding values for all individuals are computed as.

```
> L.all <- t( chol( A ) )
> BLUP <- L.all%%mod1$ranef
```

Note that the dimension of L is 43x51 and $L.all$ has the dimension 51x51.

4. EXERCISE 4

This exercise is optional. Send me an email if you want me to check your solution or if you have any question. The following code gives the results.

```
> library(MCMCglmm)
> data(BTped, package="MCMCglmm")
> head(BTped)
  animal dam sire
1 R187557 <NA> <NA>
2 R187559 <NA> <NA>
3 R187568 <NA> <NA>
4 R187518 <NA> <NA>
5 R187528 <NA> <NA>
6 R187945 <NA> <NA>
> tail(BTped)
```

```

      animal      dam      sire
1035 R187311    Fem20 R187600
1036 R187302 K983388 R187597
1037 R187732 R186903 Male111
1038 R187073 R187535 R187534
1039 R187905 R186911 Male103
1040 R187002 R187001 R187533
> data(BTdata, package="MCMCglmm")
> head(BTdata)
      tarsus      back animal      dam fosternest hatchdate sex
1 -1.89229718  1.1464212 R187142 R187557      F2102 -0.6874021 Fem
2  1.13610981 -0.7596521 R187154 R187559      F1902 -0.6874021 Male
3  0.98468946  0.1449373 R187341 R187568      A602 -0.4279814 Male
4  0.37900806  0.2555847 R046169 R187518      A1302 -1.4656641 Male
5 -0.07525299 -0.3006992 R046161 R187528      A2602 -1.4656641 Fem
6 -1.13519543  1.5577219 R187409 R187945      C2302  0.3502805 Fem
> summary(BTdata)
      tarsus      back      animal      dam
Min.   :-3.86076   Min.   :-2.71547   R046110:  1   R187926: 15
1st Qu.: -0.68093   1st Qu.: -0.69795   R046111:  1   R187517: 13
Median :  0.07617   Median : -0.06672   R046112:  1   R187518: 12
Mean    :  0.00000   Mean    :  0.00000   R046113:  1   R187559: 12
3rd Qu.:  0.53043   3rd Qu.:  0.68921   R046114:  1   Fem20   : 11
Max.    :  4.61878   Max.    :  3.47181   R046115:  1   R187030: 11
                                (Other):822   (Other):754

      fosternest hatchdate      sex
G402   : 15   Min.   :-1.9845   Fem :373
A2602   : 12   1st Qu.: -0.6874   Male:408
A602    : 12   Median : -0.1686   UNK  : 47
F2102   : 12   Mean    :  0.0000
A1602   : 11   3rd Qu.:  0.6097
B202    : 11   Max.    :  2.4256
(Other):755
> invA <- inverseA(BTped)$Ainv
> A <- solve(invA)
> image(A)
> L <- t( chol(as.matrix(A)) )
> obs.tarsus = 213:1040
> ord.data <- merge(BTdata, BTped[obs.tarsus,], sort=FALSE)
> Z0 <- diag(1040)[obs.tarsus,]
> Z <- Z0 %*% L

```

```
> y <- ord.data$tarsus
> X <- matrix(1,length(y),1)
> mod.tarsus <- hglm(X,y,Z)
```

NOTE: You are fitting a model with one Gaussian random effect term, and the number of effects (p) is greater than the number of observations (n). Consider turning on the argument 'bigRR' that may speed up a lot if $p \gg n$.

```
> plot(mod.tarsus)
> print(mod.tarsus)
```

Call:

```
hglm.default(X = X, y = y, Z = Z)
```

```
-----
Estimates of the mean model
-----
```

Fixed effects:

```
      X.1
-0.006948166
```

Random effects:

```
      Z.1      Z.2      Z.3
-0.9960177  0.5618590 -0.2392628
```

...

```
      Z.1039      Z.1040
0.5875371 0.4746152
```

NOTE: to show all the random effects estimates, use `print(hglm.object, print.ra`

Dispersion parameter for the mean model: 0.4946119

Dispersion parameter for the random effects: 0.5006175

Estimation converged in 26 iterations

!! Observation 743 is too influential! Estimates are likely unreliable !!

The heritability is $0.5/(0.5+0.495)$

5. EXERCISE 5

First we run the code given in the exercise to load and sort the data.

```

> library(SemiPar)
> data(fossil)
> #?fossil
> plot(x=fossil$age, y=fossil$strontium.ratio)
> x <- as.numeric(sort(fossil$age))
> y <- scale(fossil$strontium.ratio[sort(fossil$age, index.return=TRUE)$ix])
> plot(x,y, xlab="Age", ylab="Scaled Strontium Ratio")

```

Then we run the code for the function that produces the spatial covariance matrix. Here d is the distance matrix (i.e. the distance between all pairwise x values), l is the range parameter and ν is a smoothing parameter.

```

> ##Function to construct the Matern covariance matrix
> get.matern <-function(d, l, nu) {
+ #Distance matrix d
+ #Range l
+ #Dimension parameter nu
+ if (nu == 0.5) c <- exp(-d*l^(-1))
+ if (nu == 1.5) c <- (1+sqrt(3)*d*l^(-1))*exp(-sqrt(3)*d*l^(-1))
+ if (nu == 2.5) c <- (1+(sqrt(5)*d)*l^(-1)+(5*d^2)*3^(-1)*l^(-2))*exp(-sqrt(5)*d*l^(-1))
+ return(c)
+ }
> ##
>
> #see Ranjan et al (2011) Technometrics, 53:4, 366-378
> bend <- function(C, OK.cond = 5) {
+ eigVal <- eigen(C)$values
+ #plot(-log10(eigVal))
+ cond0 <- log10(max(eigVal)/min(eigVal))
+ delta = max(eigVal)*(10^cond0-10^OK.cond)/(10^cond0*(10^OK.cond-1))
+ C1 <- C
+ diag(C1) = diag(C) + delta
+ return(C1)
+ }

```

Now we are ready to fit the model using the hglm package.

```

> distMat = as.matrix(dist(x))
> rho = 9.1 #A good first value to try is sd(x)
> nu = 1.5 #Can be either 0.5, 1.5 or 2.5
> C1 <- get.matern(distMat, rho, nu)
> C1 <- bend(C1)
> image(C1)
> Z <- t(chol(C1))

```

```
> N <- length(y)
> X <- matrix(1, N, 1)
> library(hglm)
> mod1 <- hglm(y=y, X=X, Z=Z)
> plot(x,y, xlab="Age", ylab="Scaled Strontium Ratio")
> points(x , mod1$fv, col="red", type="l")
```

You can solve this exercise by plotting the results for different values of l and ν , and try to see which values that seem reasonable. A more sophisticated alternative is to use some cross-validation technique and maximise the predicted mean square error. Alternatively, you could maximize the REML likelihood by running the model for different values of l and ν . The model with the largest REML value is the one to prefer. The cross-validation technique and the maximum REML likelihood technique should give similar results. The maximum REML likelihood technique for Matern covariance structures is planned to be included in the hglm package in the near future.

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